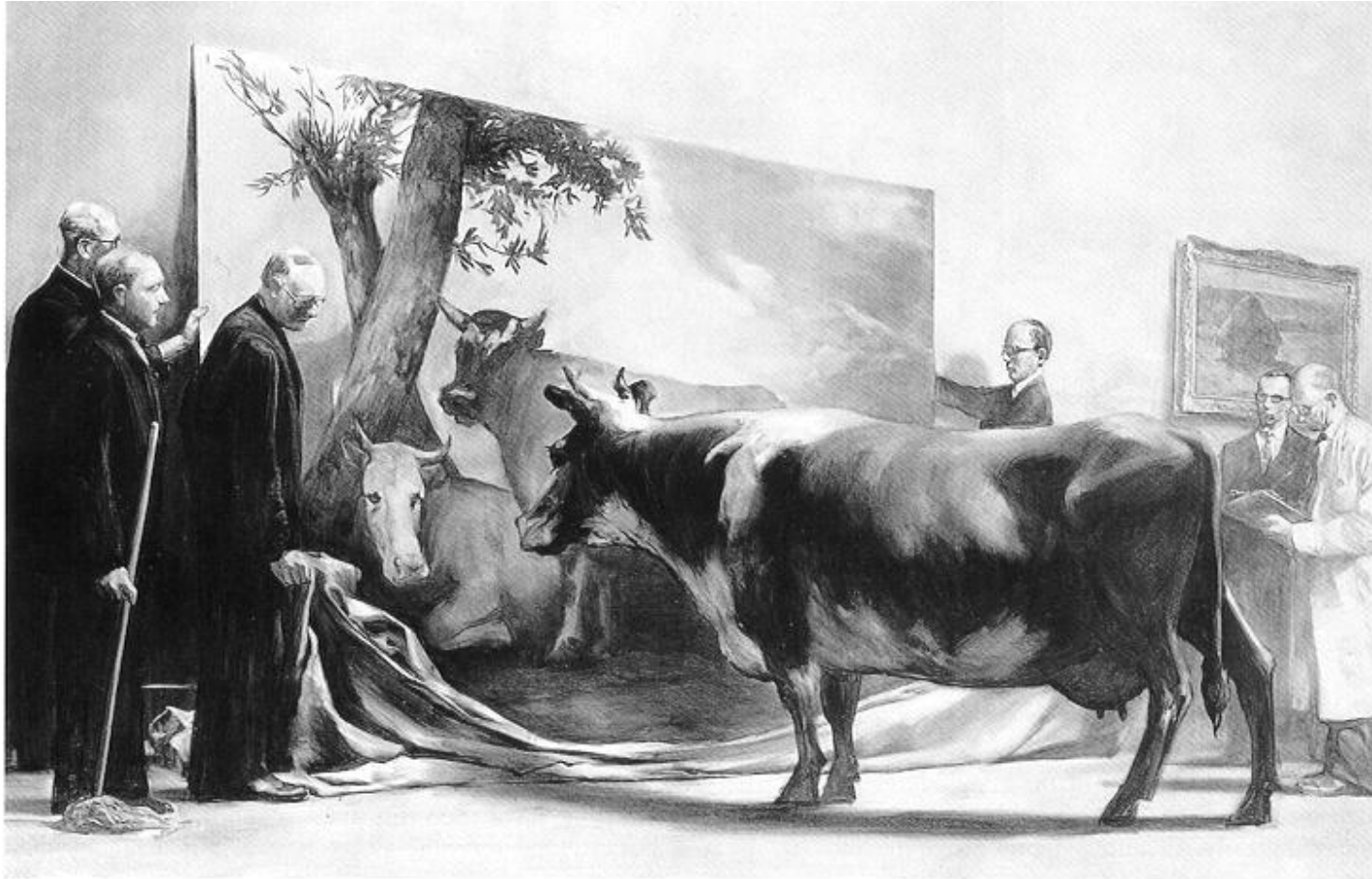


ENVX2001 Applied Statistical Methods



Lecture 2 – PCA and FA
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Outline

- Multivariate data
- Data exploration and hypothesis testing
- Important techniques including
 - PCA*
 - Cluster analysis
 - nMDS
 - Permutational approaches
 - MANOVA

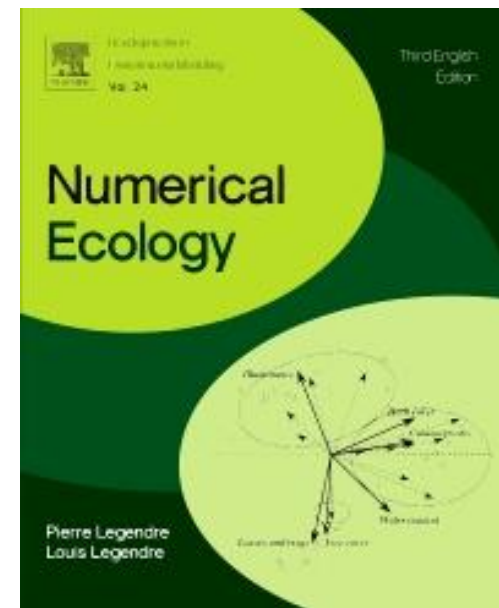
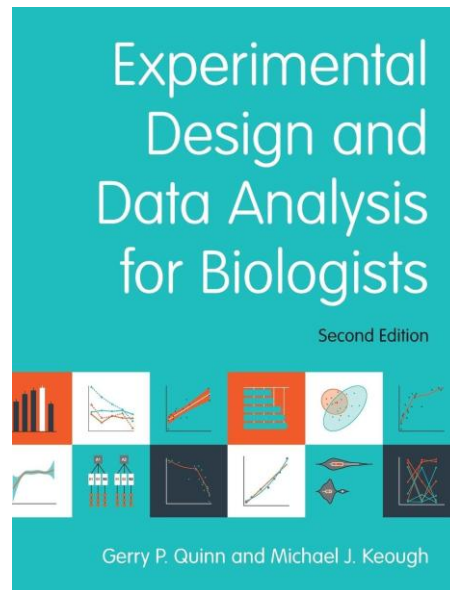
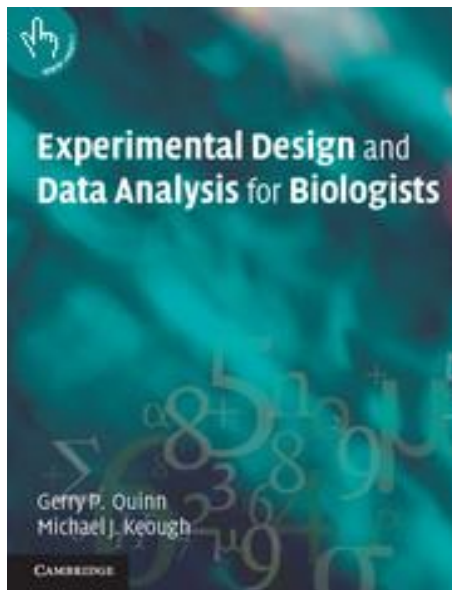
This lecture

- Principal components analysis (PCA)
- Factor Analysis - rotations
- Case study – beer
- Process for interpreting outputs



References

- Quinn and Keough (2002) – Chapter 17
- Quinn and Keough (2023) – Chapter 15
- Legendre & Legendre (2012) –Chapter 9
- Han *et al.* (2023)



First steps v1 – linear combinations

- Linear combinations of variables that summarise variation in dataset, deriving new variables
 - Principal components
 - Discriminant functions
 - Canonical functions
 - Factors
- Extracted in order – first explains most variance, **second is uncorrelated with the first** and explains most of the remaining variance, **third is not correlated with first or second** etc.
- Early consolidation of variances in derived variables

Linear combinations

- **Eigenvalues** – amount of original variances explained by new derived variable (usually expressed as a %)
- **Eigenvectors** – the directions (axes) in variable space. Each component is a linear combination of the original variables with these weights.
- Matrix algebra underpins PCA, but **no matrix operations are expected in this course** — we focus on interpretation.
- PCA assumes **linearity** among variables.

Goals of PCA

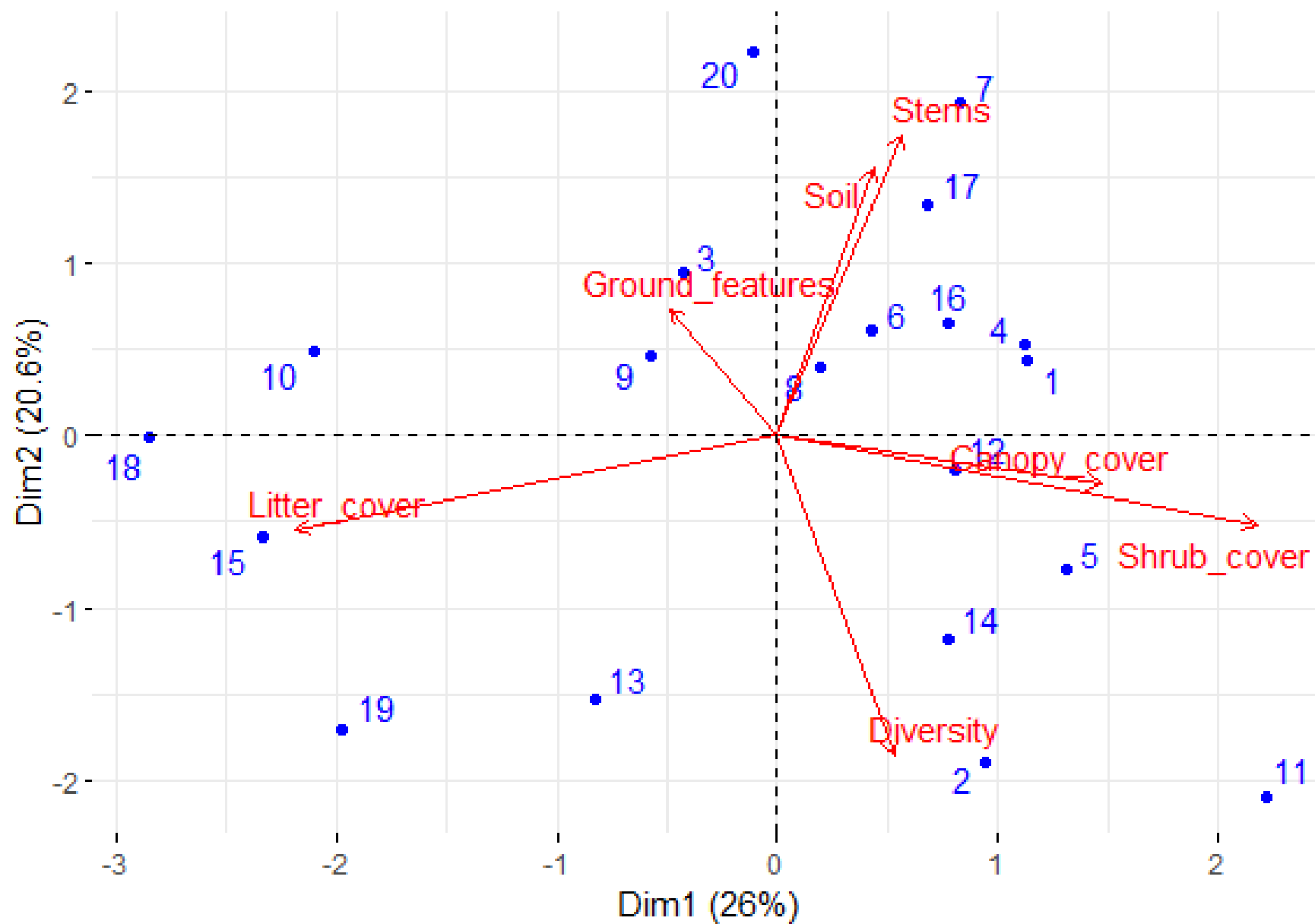
- **Identify latent variables**
(e.g., *habitat complexity*, *fitness*) — underlying constructs that explain patterns in observed variables
- **Reduce dimensionality**
Simplify data by capturing the most important variation in fewer variables
- **Summarise correlation structure**
Understand how variables relate to each other through shared components or factors
- **Develop or test theory and instruments**
Inform the design of scales, indices, or theoretical models

Habitat complexity in forests

- Rumored to be important in driving diversity (species richness, community composition etc.)
- What can we measure?
 - Canopy cover
 - Shrub cover
 - Litter (ground) cover
 - Numbers of stems
 - Diversity (trees, shrubs, understory)
 - Ground characteristics (logs, rocks, dirt cover)
 - Soil characters (moisture, nutrients)



PCA - Biplot

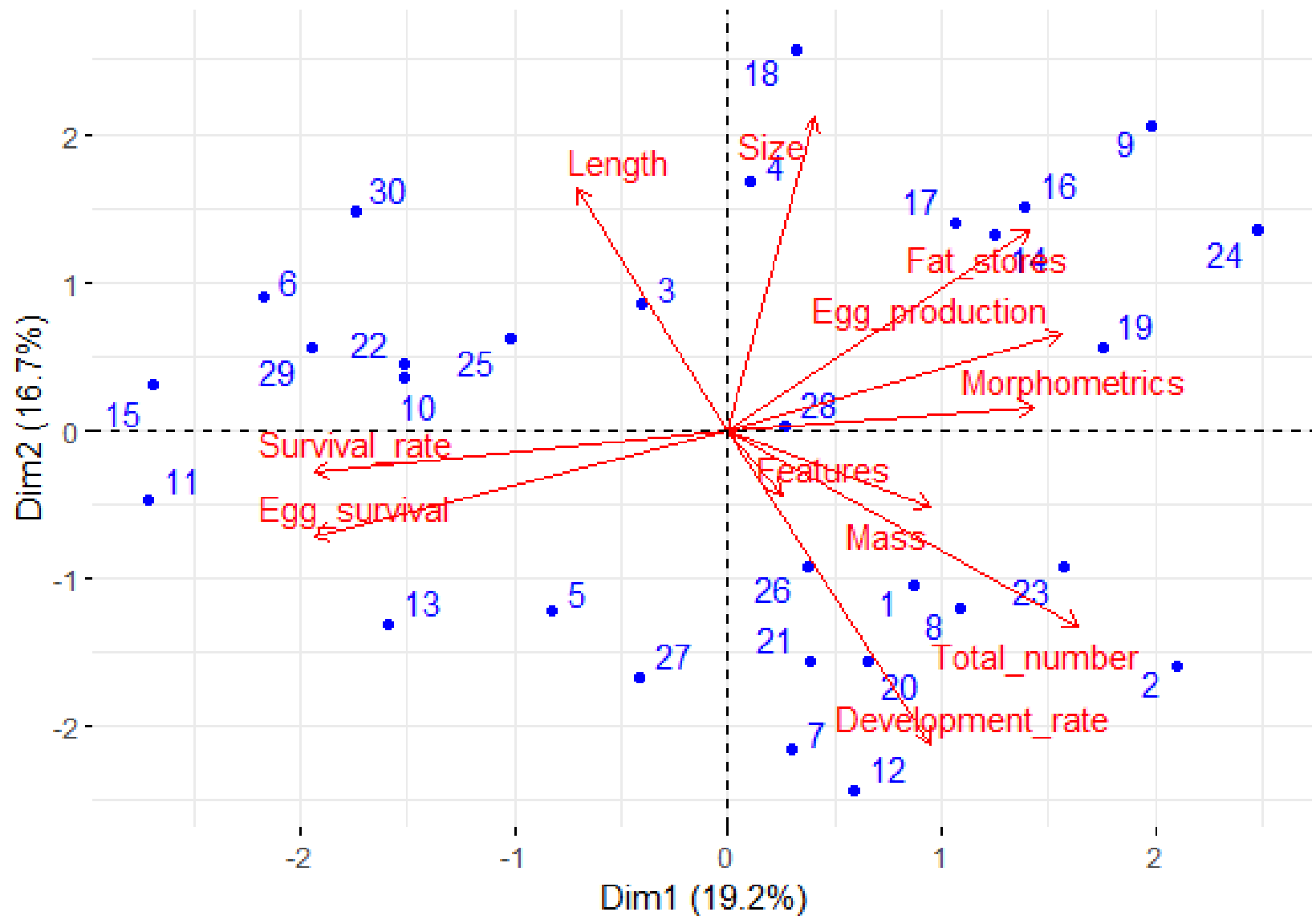


Fitness?

- Mass
- Fat stores
- Morphometrics
 - Length(s)
 - Features of significance (wings, legs, ovipositors etc.)
- Survival rates
- Development rates
- Egg production
 - Size
 - Total number
- Egg survival

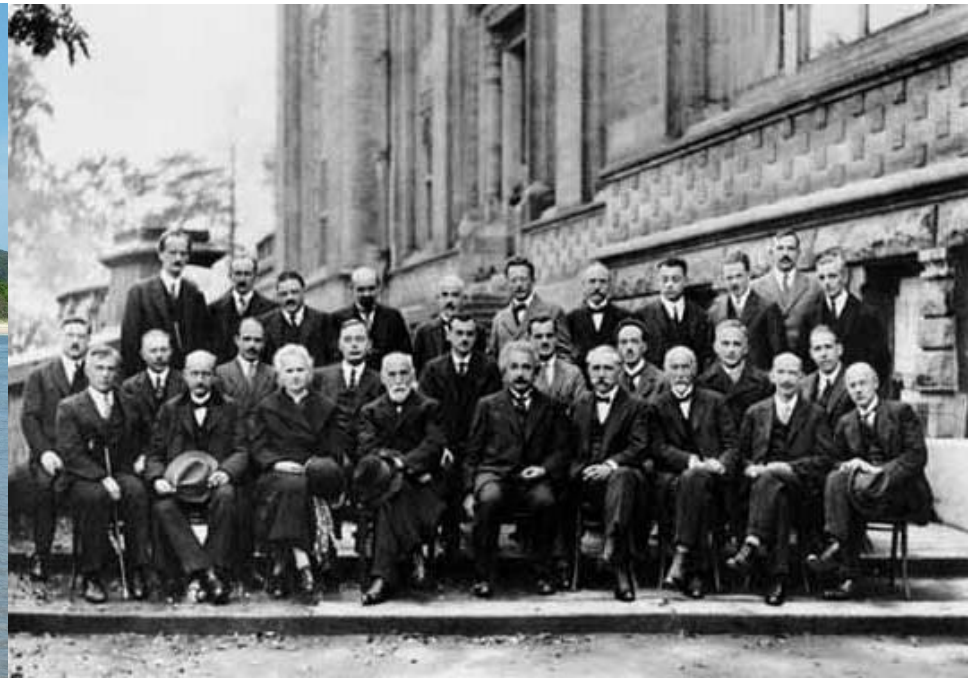


PCA - Biplot



Collecting lots of data

Data exploration: fishing expedition or hypothesis driven?



Factors (not the ANOVA kind!)

- Correlations between pairs of variables
 - R- matrix (collection of Pearson correlation coefficients - r)
- Diagonals all 1
- Factor analysis reduces the matrix to its underlying dimensions – looking for possible clusters of correlations
- Here is the correlation matrix for the iris data in R

```
>data(iris)
>cor(iris[,1,4])
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
Sepal.Length	1	-0.117569784	0.871753776	0.817941126
Sepal.Width	-0.117569784	1	-0.428440104	-0.366125933
Petal.Length	0.871753776	-0.428440104	1	0.962865431
Petal.Width	0.817941126	-0.366125933	0.962865431	1

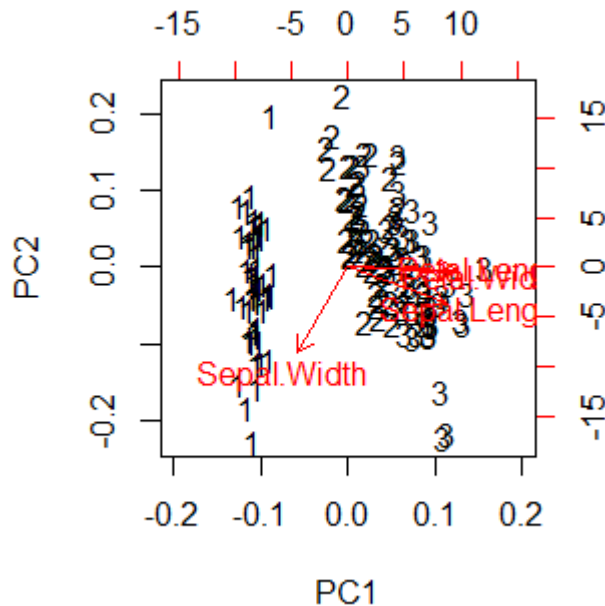
Principal Components Analysis

- Identify **underlying dimensions** or principal components within a dataset
- Helps understand the joint or common variation among a set of variables
- Probably the most commonly used method of deriving “factors” in factor analysis (before rotation)
- PCA is a **variable-reduction technique**, while FA aims to model **latent variables causing observed correlations**.

PCA - the question(s)

- Can we reduce variables down based on their **linear relationships**?
- Can we generate new ways of describing complex datasets?
- Are there **intercorrelations** between variables?
- Can we identify **latent (underlying) variables** driving the observed patterns?

PCA - the R output



Importance of Components

	PC1	PC2	PC3	PC4
Standard deviation	1.7084	0.956	0.38309	0.14393
Proportion of Variance	0.7296	0.2285	0.03669	0.00518
Cumulative Proportion	0.7296	0.9581	0.99482	1

Loadings (rotations)

	PC1	PC2	PC3	PC4
Sepal.Length	0.5210659	-0.37741762	0.7195664	0.2612863
Sepal.Width	-0.2693474	-0.92329566	-0.2443818	-0.1235096
Petal.Length	0.5804131	-0.02449161	-0.1421264	-0.8014492
Petal.Width	0.5648565	-0.06694199	-0.6342727	0.5235971

PCA - the process/meaning

- Explore **associations** between variables and **generate hypotheses**
- **Simplify complex datasets** and create new composite variables (principal components)
- Extract **PC scores** for each observation and use them in:
 - **Regression models**
 - **ANOVA** (e.g., comparing “habitat complexity” across management types)
- Example:
Use a **PC score representing “complexity”** to predict **species richness** at different sites

Principal Components

- **PC1:** Direction capturing the greatest variation in the data
- **PC2:** Perpendicular to PC1; captures the next highest remaining variation
- **PC3:** Perpendicular to PC1 and PC2; captures remaining variation
- Additional components follow the same pattern
- Each component is a linear combination of the original variables

When to Use PCA

- You have **p continuous variables**
- These variables are often **correlated**
- You want to reduce them to **fewer variables (m components)** where **$m < p$**
- Each component captures a **different pattern of variation** in the data

Covariance or Correlation Matrix

- If units of x and y are the same (i.e. temperature) or with similar orders or magnitude, use a covariance matrix
 - May need to standardise units
- If units are different, and/or orders of magnitude are different, use a correlation matrix as it standardises the units
- **Standardising** a covariance-based analysis (i.e., converting variables to z-scores) effectively **turns it into a correlation-based analysis**

Components and Variables

- Each component is a weighted linear combination of the variables (Eigenvectors)

$$C_i = W_{i1}X_1 + W_{i2}X_2 + \dots + W_{ip}X_p$$

- Each variable is a weighted linear combination of the components.

$$X_j = A_{1j}C_1 + A_{2j}C_2 + \dots + A_{mj}C_m$$

A Simple, Contrived Example

- Consumers rate importance of seven characteristics of beer;
 - Low cost
 - High size of bottle
 - High alcohol content
 - Reputation of brand
 - Color
 - Aroma
 - Taste



Source: Karl Wuensch
Department of Psychology, East Carolina University

Extracting Principal Components

- From p variables we can extract p components.
- Each of p **eigenvalues** represents the amount of standardized variance that has been captured by one component
- The first component accounts for the largest possible amount of variance
- The second captures as much as possible of what is left over, and so on
- Each is orthogonal to the others

Properties of PCA

- Each variable is **standardised** $\rightarrow$ variance = 1
- Total variance across all **p variables** = p
- PCA redistributes this variance across **p components**
- The **sum of all eigenvalues** = p
- **No variance is lost** (it is just reorganised)
- Variance explained by a component = **eigenvalue / p**

Let's do PCA in R

```
pca_Beer <- prcomp(beer[,1:7],  
scale = T)  
#This does the principal  
components and scales them  
screeplot(pca_Beer, type =  
"lines", main = "Screeplot Beer  
PCA")  
#This is the screeplot  
summary(pca_Beer)
```

- Beer data - eigenvalues and proportions of variance for the seven components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	1.8201	1.6173	0.75804	0.48978	0.36668	0.29231	0.19206
Proportion of Variance	0.4733	0.3737	0.08209	0.03427	0.01921	0.01221	0.00527
Cumulative Proportion	0.4733	0.847	0.92905	0.96332	0.98252	0.99473	1

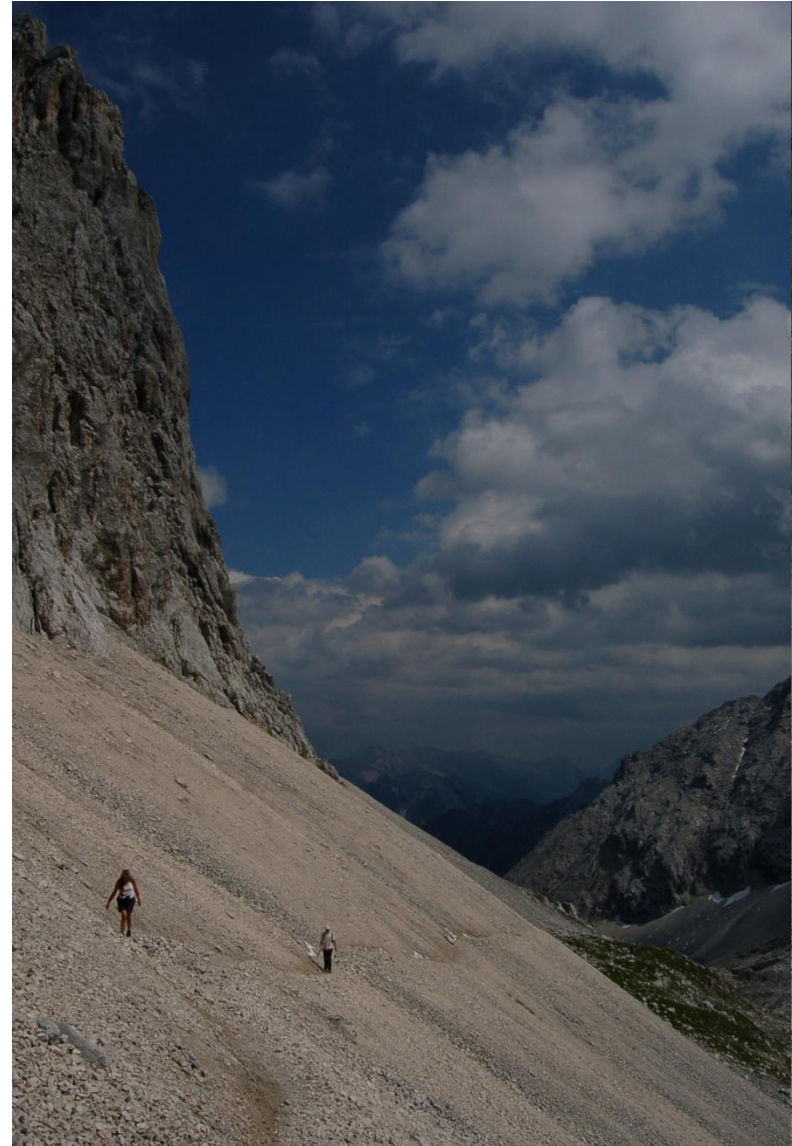
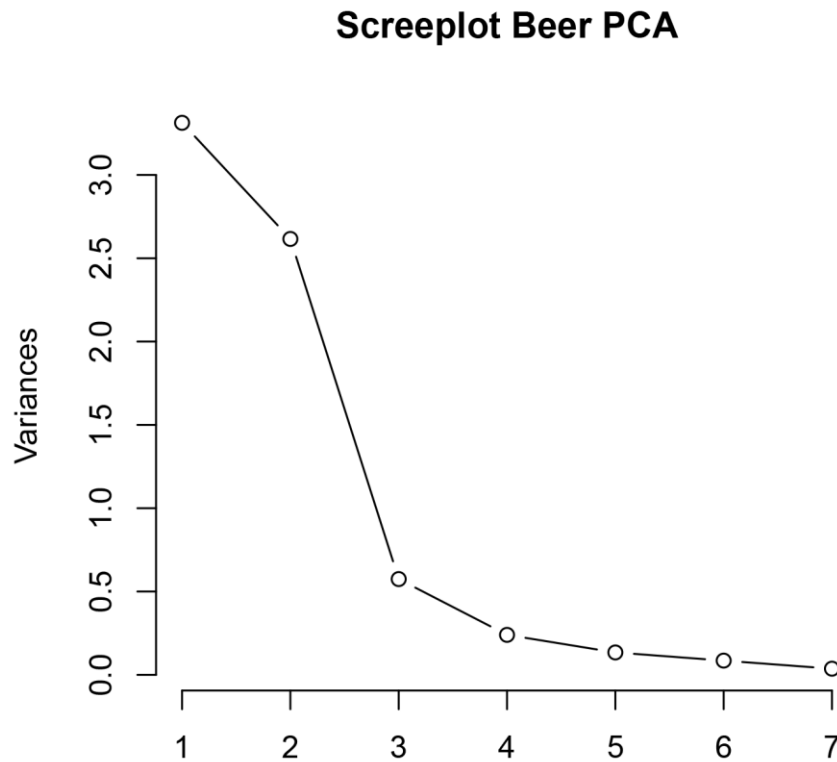
Remember Eigenvalue = (Standard deviation)²

NB: problems with unique variables –check correlation matrix

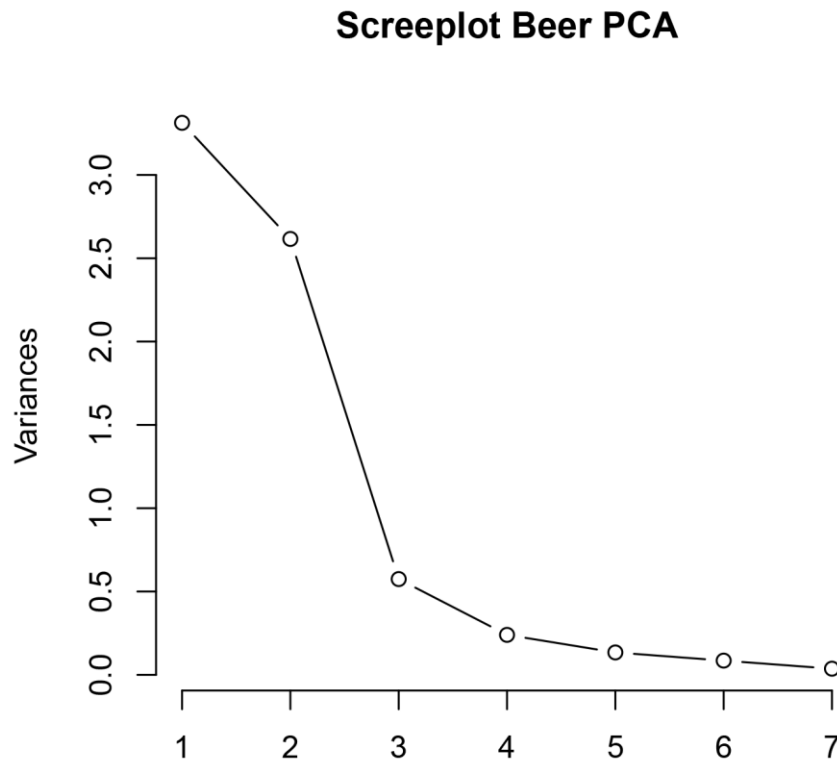
How many components to retain?

- **How Many Components to Retain?**
 - From **p variables**, you can extract **p components**
→ but we usually keep **fewer ($m < p$)**
- **Rule of Thumb (Kaiser's Criterion)**
 - Keep components with **eigenvalues ≥ 1**
- **Why?**
 - An eigenvalue represents **variance explained**
 - A component with eigenvalue < 1 explains **less variance than a single original variable**
→ usually not worth keeping
- **Key Reminder**
 - **Eigenvalue = variance = (standard deviation)²**

- Visual Aid: Use a Scree Plot
- For the beer data



- Visual Aid: Use a Scree Plot
- For the beer data

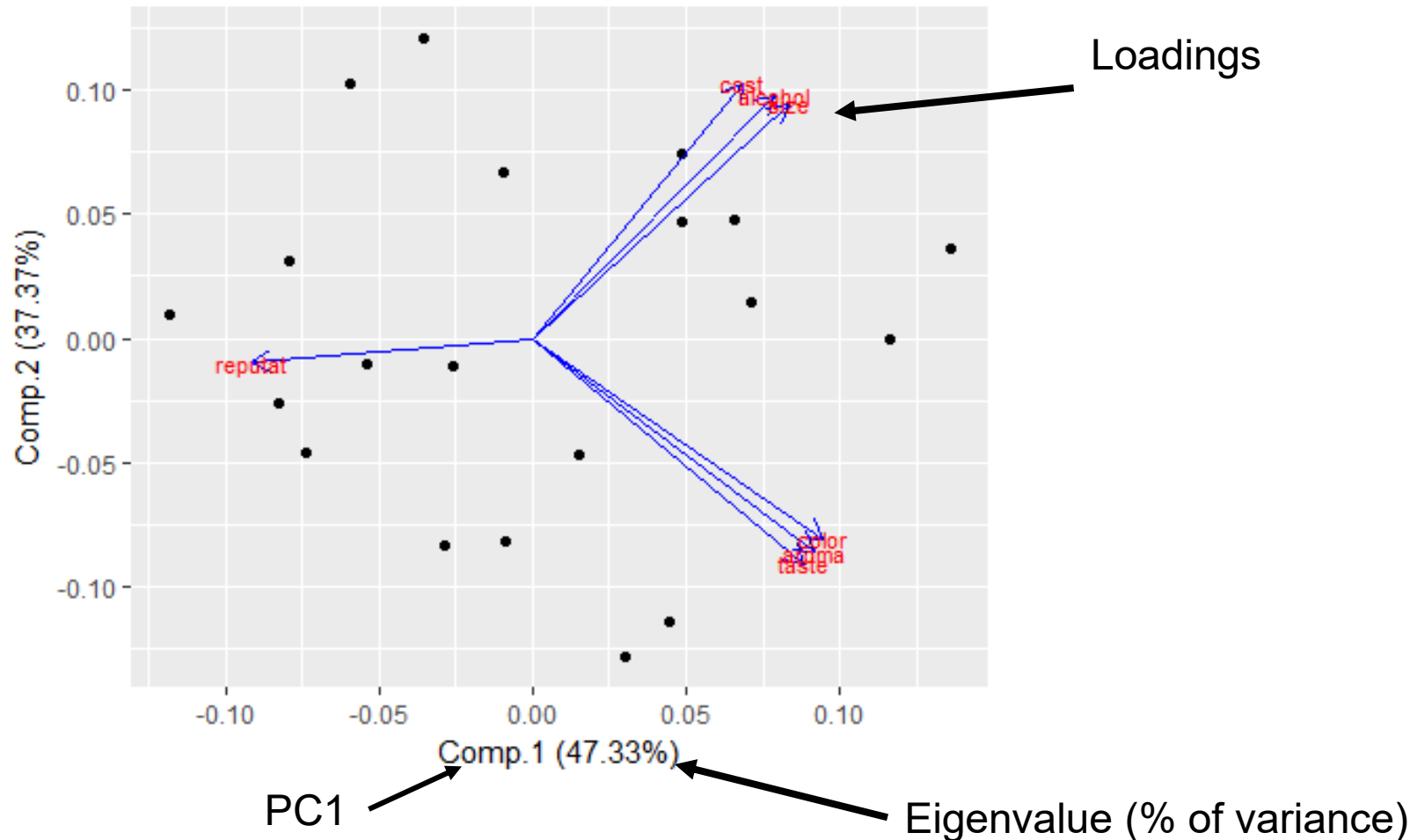


- Only the first two components have eigenvalues greater than 1
- Big drop in eigenvalue between component 2 and component 3
- Components 3-7 are scree.
- 2 component solution.
- Can look at solution with one fewer and with one more component.

Loadings and factor rotation

- loading matrix = factor pattern matrix = component matrix = rotations (in R)
- Often shown as vectors
- Each loading is the **Pearson r** correlation between one variable and one component
- Most variables will tend to have high loadings on the first factors and small loadings on the others
- Rotation helps discriminate among factors
- Unrotated loadings for our 2 component solution:

Visualising PCA with Biplots

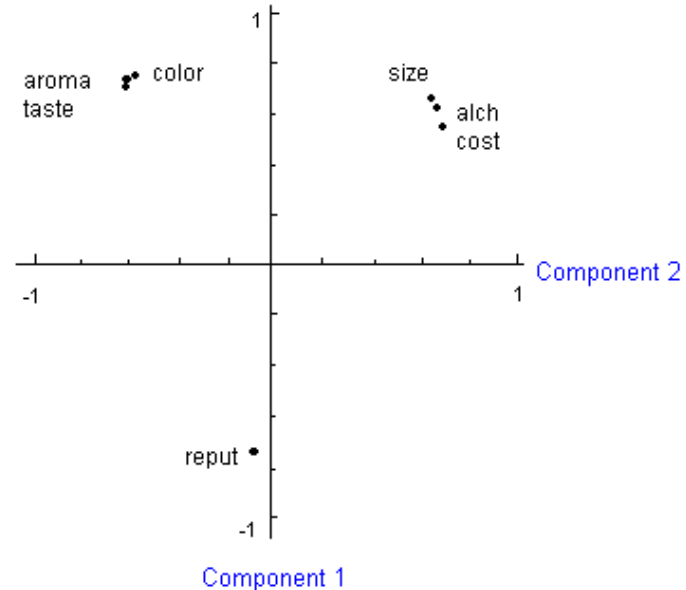


Component Matrix^a

	Component	
	1	2
COLOR	.760	-.576
AROMA	.736	-.614
REPUTAT	-.735	-.071
TASTE	.710	-.646
COST	.550	.734
ALCOHOL	.632	.699
SIZE	.667	.675

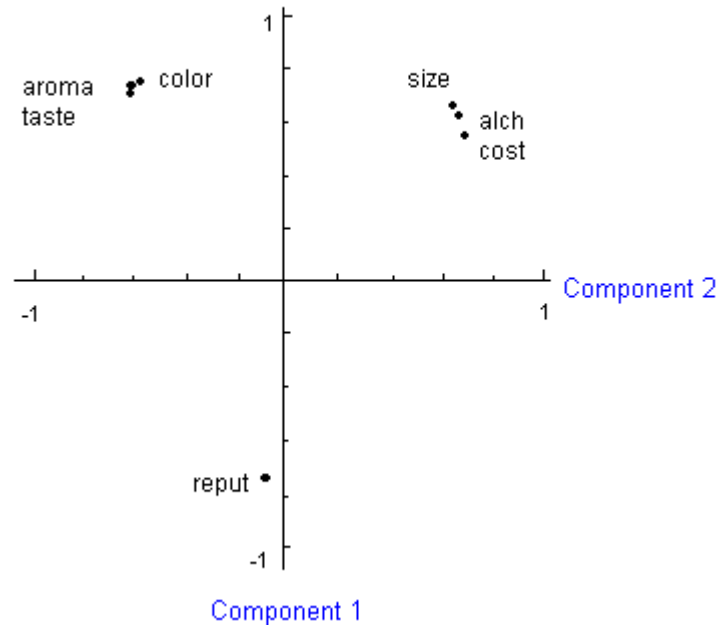
Extraction Method: Principal Component Analysis.

a. 2 components extracted.

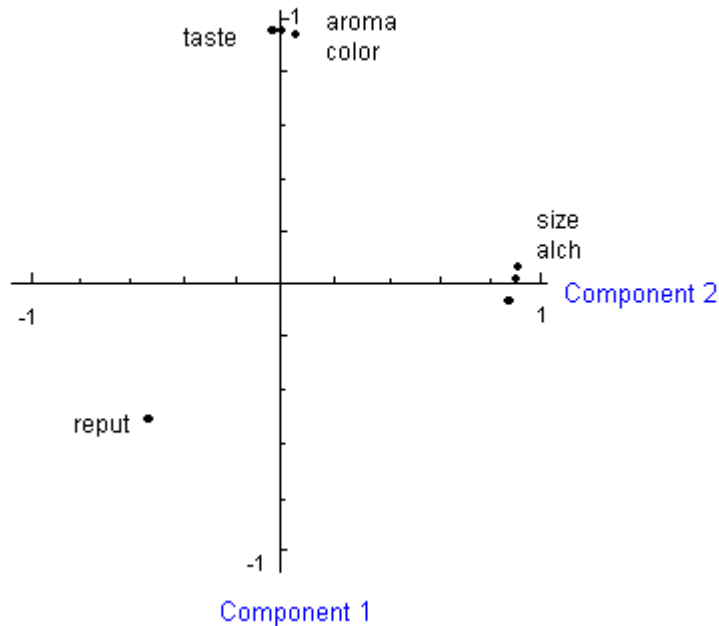


- All variables load well on first component, economy and quality vs. reputation
- Second component is more interesting... economy versus quality?

Rotate axes so that the two dimensions pass more nearly through the two major clusters (COST, SIZE, ALCH and COLOR, AROMA, TASTE)



Rotate axes so that the two dimensions pass more nearly through the two major clusters (COST, SIZE, ALCH and COLOR, AROMA, TASTE)



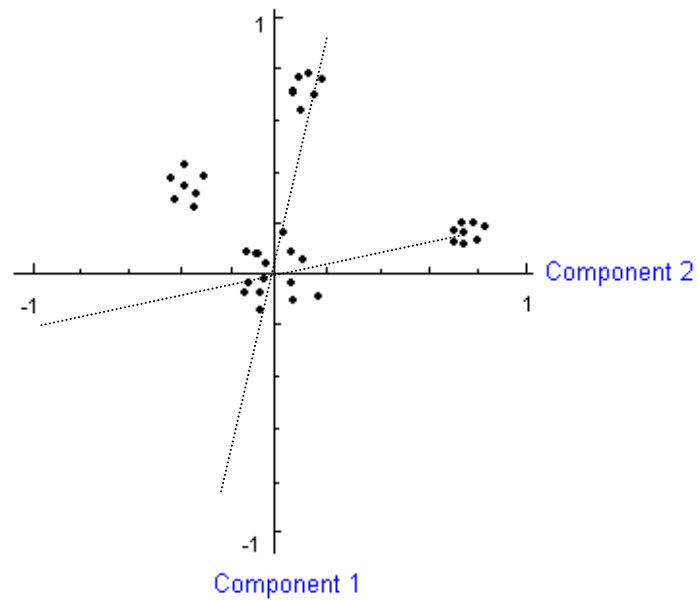


Rotating

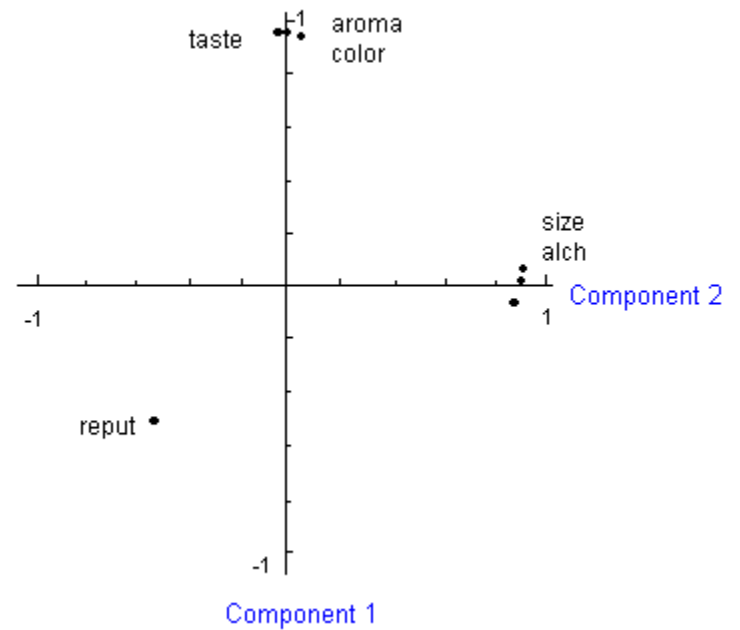


- Done as type of Factor Analysis
 - Range of options
 - Orthogonal vs oblique
 - Keep independent vs permit correlations
 - Orthogonal options
 - **Quartimax** (maximise spread of loadings, large loadings larger and small loadings smaller within each **variable**) – not used much in R
 - **Varimax** (maximise dispersion of loadings, large loadings larger and the small loadings smaller within each **component**)
- Promax** (an oblique Varimax)

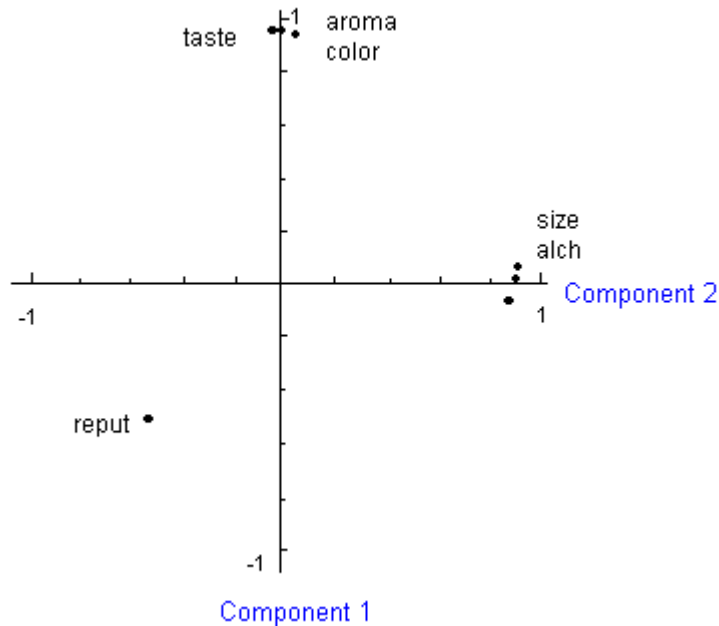
Oblique



Orthogonal



Rotate axes so that the two dimensions pass more nearly through the two major clusters (COST, SIZE, ALCH and COLOR, AROMA, TASTE)



Let's do a Varimax rotation in R

```
loadings_Beer <- loadings(pca_Beer,  
cutoff=0.5, Sort=TRUE)  
varimax((loadings_Beer), normalize =  
FALSE)  
# OR Through the Factor Analysis  
command  
fa1 <- factanal(beer, 3,  
rotation="varimax")  
fa1
```

Naming Components

	Factor1	Factor2
cost		0.839
size		0.993
alcohol		0.904
reputat	-0.342	-0.348
color	0.994	
aroma	0.913	
taste	0.913	

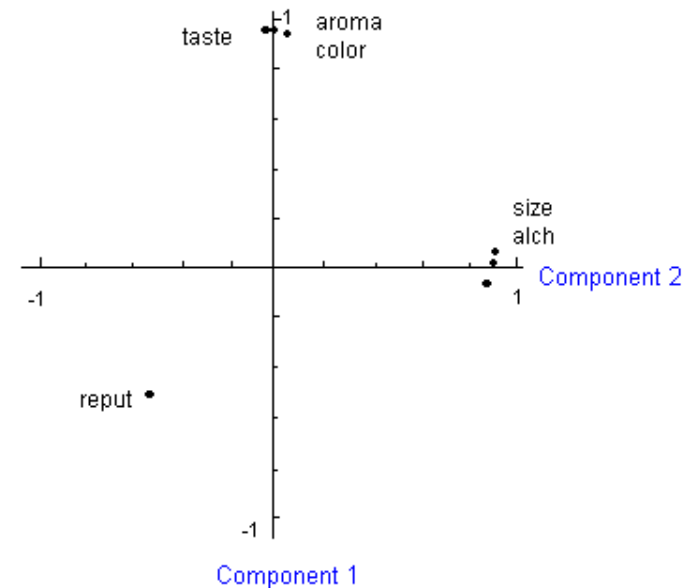
- For each component, look at how it is correlated with the variables
- Name the construct represented by that factor – important for telling the story
- If you cannot, perhaps you should try a different solution.

Our components...

“aesthetic quality” and “cheap drunk.”

- Component 1 = Quality versus reputation.
- Component 2 = Economy (or cheap drunk) versus reputation.

	Factor1	Factor2
cost		0.839
size		0.993
alcohol		0.904
reputat	-0.342	-0.348
color	0.994	
aroma	0.913	
taste	0.913	



Number of Components in the Rotated Solution

- Use **trial and refinement**:
 - Try **one fewer** and **one more** component
 - Compare which solution is **most interpretable**
- Does the solution make **biological / practical sense**?
- **Overextracting (too many components)**
 - usually safer than underextracting
 - similar to adding predictors increasing R^2
- **Underextracting (too few components)**
 - risks missing important structure in the data
- If there is only **one true underlying factor**, extracting multiple components can lead to:
 - **“Factor splitting”**
 - one real pattern artificially split into several components

Explained variance

- For each component get the amount of variance explained
- Prior to rotation, these are eigenvalues the SSL for our data, after rotation:

Total Variance Explained

Component	Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %
1	3.017	43.101	43.101
2	2.912	41.595	84.696

Extraction Method: Principal Component Analysis.

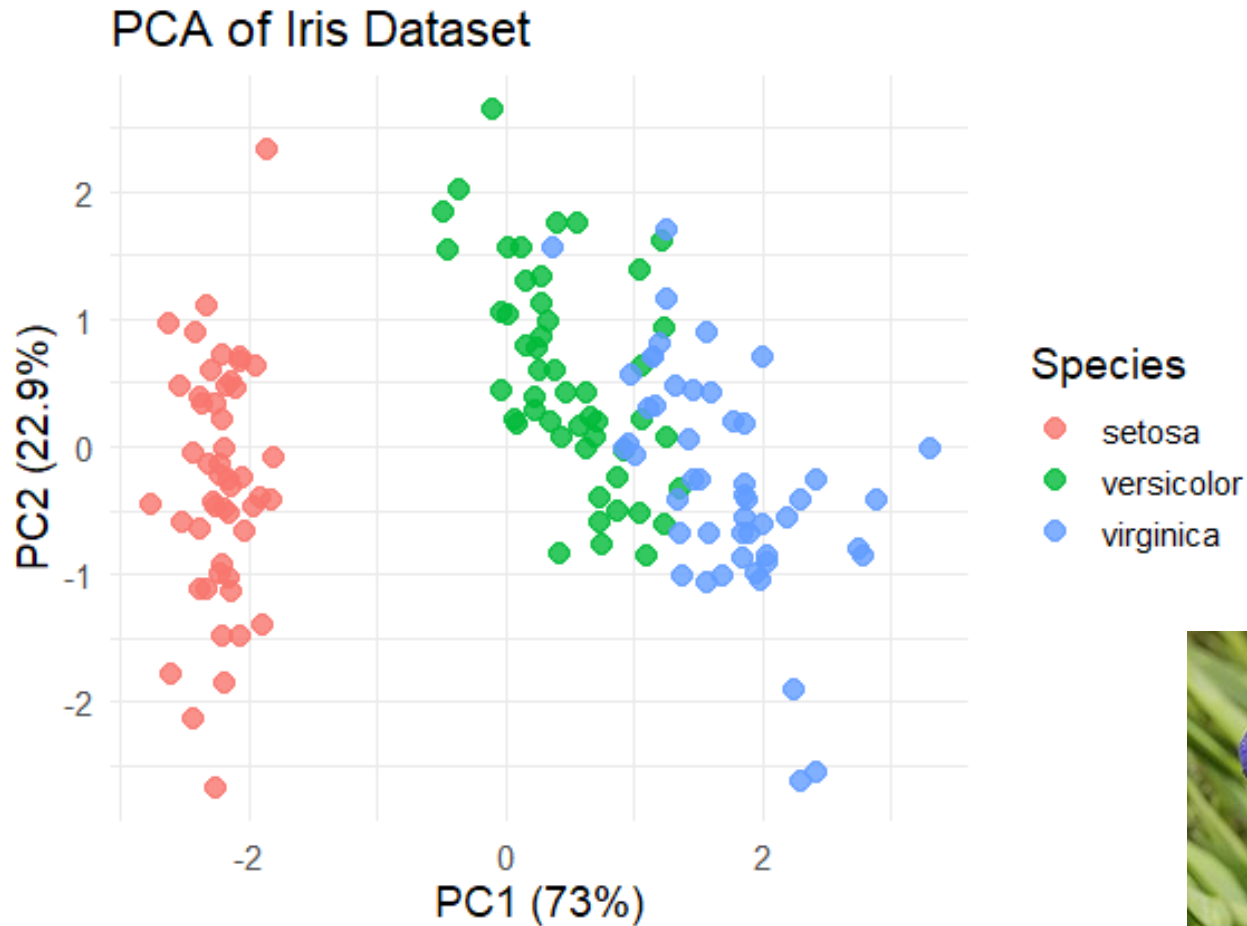
- After rotation the two components together account for $(3.02 + 2.91) / 7 = 85\%$ of the total variance.

Doing more with PCA?

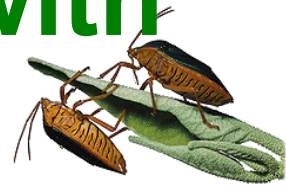
- Each original contributing replicated unit (landscapes, people) has a score for each PC
- *Can be used as predictor or response variable*



Looking for Groupings in PCA

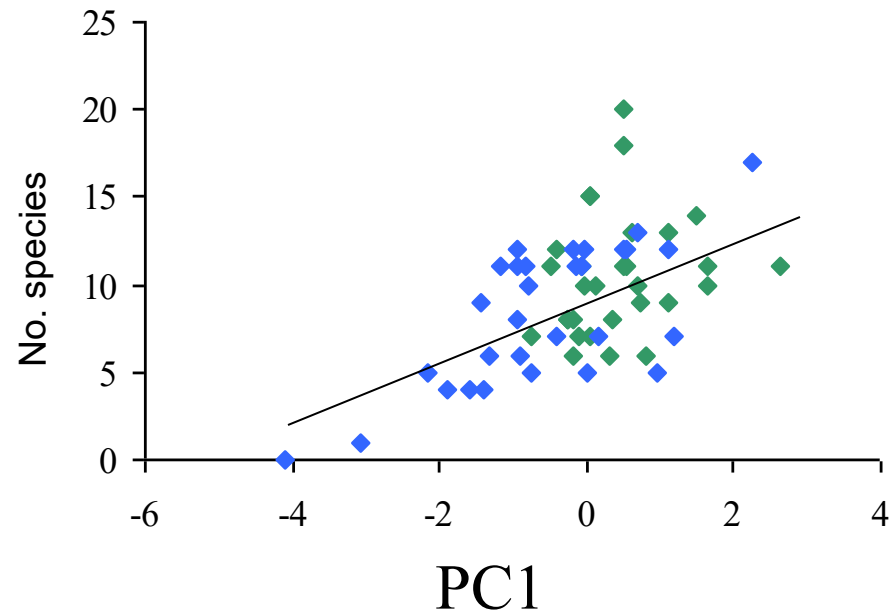


Increased species richness with vegetation complexity



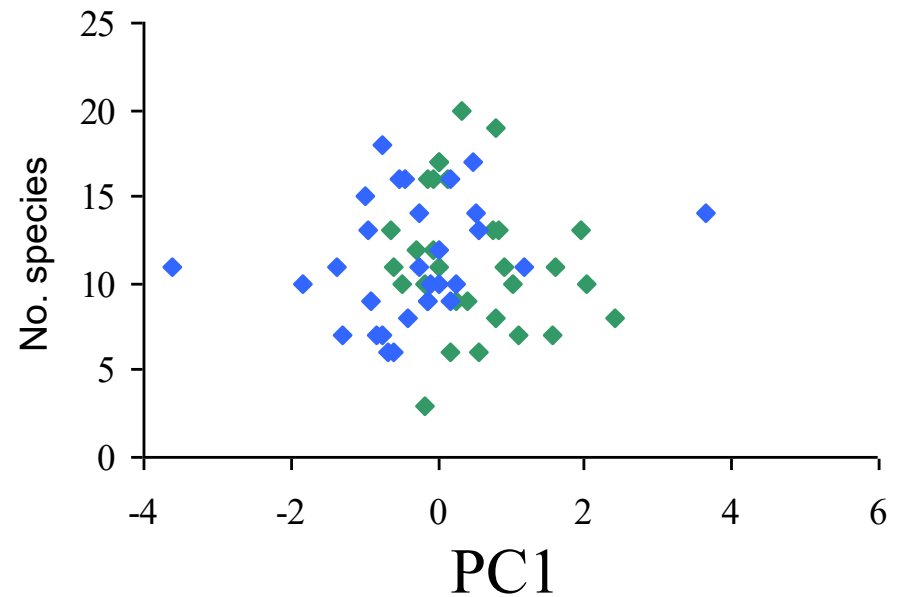
Set grazing

$R^2 = 0.34$, $p < 0.0001$, $n = 60$



Flex grazing

$R^2 = 0.05$, $p > 0.05$, $n = 60$



Complexity →

Complexity →

Thompson, D.B. 1992. Consumption rates and the evolution of diet-induced plasticity in the head morphology of *Melanoplus femurrubrum* (Orthoptera: Acrididae). *Oecologia* 89:204-213

Fed grasshoppers clover or rye to induce changes in head size

Table 5. Principal components eigenvector loadings for morphological variables in adult grasshoppers. The percentage of total phenotypic variance explained is % V

Morphology	PC1	PC2
Anterior articular width	0.391	0.134
Posterior articular width	0.393	0.116
Mandible depth	0.364	0.350
Head length	0.351	0.493
Head width	0.398	0.024
Femur	0.374	-0.525
Tibia	0.370	-0.571
% V	86	7

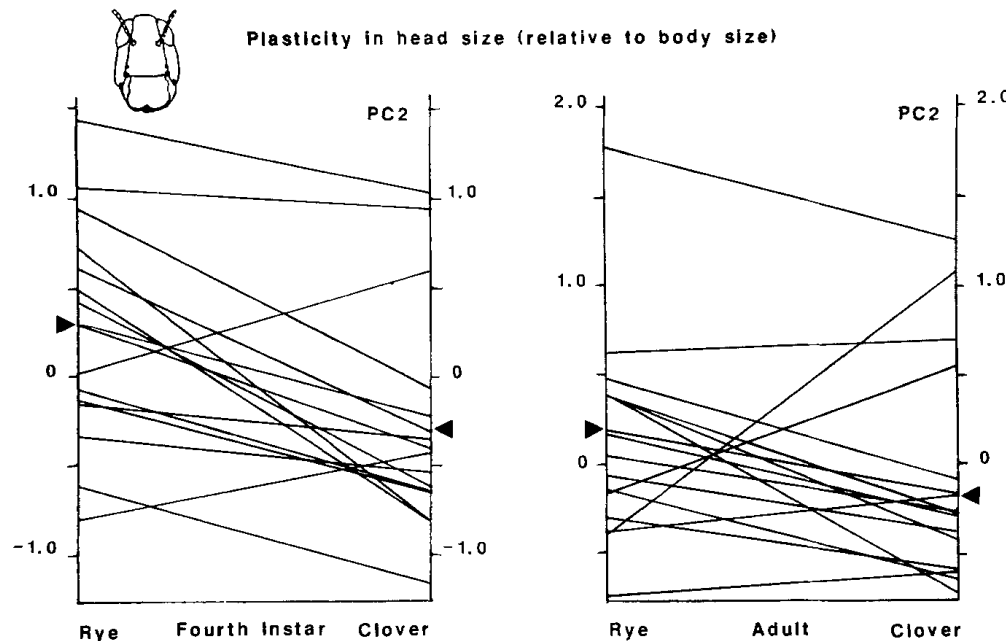


Fig. 1. Phenotypic plasticity in principal component 2 (head size relative to body size) for fourth instar and adult *M. femurrubrum* reared on rye grass or clover diets. Each line connects the means of a full-sib family and arrows indicate the phenotypic mean in each plant environment

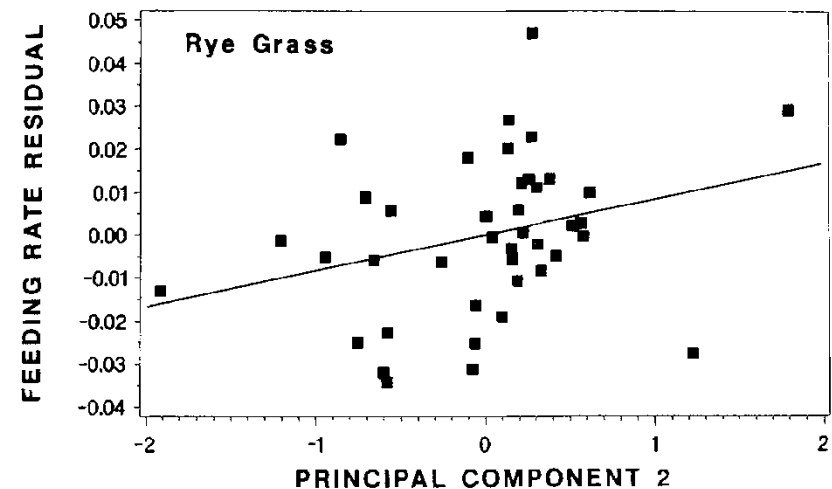


Fig. 3. Regression of consumption rate residuals (see text) on Principal Component 2 (relative head size) for grasshoppers raised on a rye grass diet when they were feeding on rye grass ($y=0.0115x+0.0013$; $P=0.01$; $r^2=0.16$)

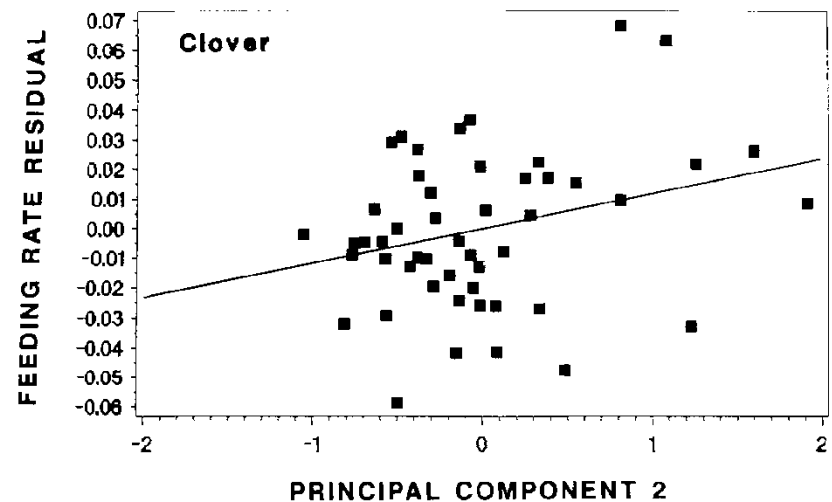


Fig. 4. Regression of consumption rate residuals (see text) on Principal Component 2 (relative head size) for grasshoppers raised on a clover diet when they were feeding on clover ($y=0.011x+0.005$; $P=0.04$; $r^2=0.11$)

PCA - summary

- Subjectivity – arbitrary rules inherent in all stats
- Correlation matrix
- Discretionary decisions
 - Cut-off for factors
 - Rotation type
 - Factor naming
 - Sample sizes
 - Biological or Environmental Meaning?

